

UMESH KANNA K B

+91 6383043548 | umeshec585@gmail.com | LinkedIn | GitHub | HDLBits Stats | Chennai

PROFESSIONAL SUMMARY

Electronics and Communication Engineer with experience in digital hardware design, RTL development, and FPGA-based systems. Practical exposure to Verilog/SystemVerilog HDL, synthesis, and simulation workflows, with entry-level foundations in Computer architecture and VLSI design methodologies using CAD Tools. .

INTERNSHIPS

Integrated VLSI Design Intern

Sept 2025 – Present

SURE Trust (ProEd), AP, India (AICTE Approved) - Remote

- Undertaking comprehensive areas covering **MOS device physics, CMOS process technology, RTL design (Verilog, CDC, TCL), SystemVerilog & UVM verification, synthesis and DFT, STA, PnR, PV, analog layout design**, and **semiconductor device modeling with process simulation**, achieving high proficiency guided by industry level professionals.

EDUCATION

College of Engineering Guindy, Anna University

Oct 2022 – May 2026

Bachelor of Engineering in Electronics and Communication with current CGPA of 8.37

Chennai, Tamil Nadu

- Relevant Coursework: Digital System Design, Computer Architecture, Digital VLSI.

SKILLS

Hardware Description Languages: Verilog, SystemVerilog (RTL Design, SVA – Basics), UVM (Basics).

Programming Languages: C/C++ (Fundamentals) **Scripting:** TCL (Basics).

Tools: Xilinx Vivado, Cadence Genus, Cadence Innovus, EDA Playground.

Systems: Linux (Basics), Git (Basics). **Hardware:** FPGA(Artix-7 Platform), Oscilloscope.

Soft Skills: Problem Solving, Teamwork, Communication.

PROJECTS

1. SPI Master–Slave Communication Protocol Verification

Nov 2025

- Designed SystemVerilog RTL for SPI Master and Slave supporting **4 SPI modes (CPOL/CPHA)**.
- Verified 8-bit and 16-bit data transactions with full-duplex communication.
- Developed a 4-component modular verification environment (generator, driver, monitor, scoreboard).
- Executed 20+ directed and self-checking testcases to validate timing and protocol compliance.
- Achieved zero data mismatches across simulated transaction scenarios.

2. Physical Design of Radix-4 Booth Multiplier – RTL to GDS Flow

Apr 2025

- RTL synthesis of a 16-bit Radix-4 Booth Multiplier using Cadence Genus with timing and area analysis.
- Constraint-driven optimization and critical-path evaluation in **Cadence Genus**.
- Floorplanning, placement, and routing performed in **Cadence Innovus**.
- Identified 3 physical-design gaps (no CTS, unconstrained clocks, sign-off limits).
- Post-layout review of routing completion, utilization, and power estimates in Innovus.

3. MIPS32 Single-Cycle Processor Design

Mar 2025

- Designed a 32-bit MIPS single-cycle processor in Verilog with a unified datapath and control unit.
- Implemented 3 instruction formats (R-type, I-type, J-type) across 5 logical stages: **IF, ID, EX, MEM, and WB**.
- Developed 6+ synthesizable RTL modules including ALU, register file, program counter, instruction memory, and data memory interfaces.
- Verified functionality using directed test programs and waveform-based simulation to validate instruction execution.
- Structured the design using a modular and hierarchical RTL approach to support scalability and reuse.

4. CMOS Inverter Layout

Aug 2024

- Designed a CMOS inverter layout in 90 nm technology using **Cadence Virtuoso Layout Suite**.
- Implemented PMOS and NMOS sizing to achieve balanced rise and fall delays.
- Successfully completed 100% DRC-clean and LVS-matched layout verification.
- Simulated Voltage Transfer Characteristics (VTC) and extracted noise margins (NMH, NML).
- Calculated propagation delay (tpHL, tpLH) using post-layout simulation results.

CERTIFICATIONS

Computer Architecture and Organization - NPTEL (IIT Kharagpur), Elite - 66%

Digital IC Design - NPTEL (IIT Madras) - 57%

VLSI Physical Design with Timing Analysis – NPTEL (IIT Roorkee), Elite- 72%

EXTRACURRICULARS & ACHIEVEMENTS

Treasurer and VTools Manager for IEEE, APS, CEG, streamlining financial operations. Contributed as a Core Member (Technical) in ECEA, CEG. Led the **CEG Fabless and CTF Projects 2025** teams ensuring successful project execution. Earned Distinction in Senior and Junior Grade Typewriting examinations.